

Exercise 7: Financial Forecasting

Scenario

You are developing a financial forecasting tool that predicts future values based on past data.

1. Understand Recursive Algorithms

What is Recursion?

Recursion is a programming technique in which a method calls itself repeatedly until a base condition is reached. It is useful for solving problems that can be broken down into smaller subproblems.

A recursive function consists of:

- Base Case – Stops the recursion.
- Recursive Case – Calls the function again with a smaller or modified input.

Advantages of Recursion

- Produces simple and readable code.
- Suitable for divide-and-conquer problems.
- Reduces the complexity of solving repetitive tasks.

2. Setup

A Java project named FinancialForecasting was created.

The project contains a code folder with the following file:

FinancialForecasting

└── code

└── FinancialForecasting.java

- FinancialForecasting.java – Implements a recursive algorithm to predict future values based on a constant annual growth rate.

3. Implementation

Forecasting Method

The recursive method calculates the future value using the formula:

$$\text{Future Value} = \text{Present Value} \times (1 + \text{Growth Rate})^n$$

where:

- Present Value = Initial investment
- Growth Rate = Annual growth rate
- n = Number of years

The recursive method repeatedly multiplies the current value by the growth rate until the required number of years is reached.

4. Analysis

Time Complexity

Operation	Time Complexity
Recursive Forecast	O(n)

where n is the number of years.

Space Complexity

Operation	Space Complexity
Recursive Forecast	O(n)

The recursion stack stores one function call for each year.

Optimization

The recursive solution can be optimized using:

- Memoization to avoid repeated calculations.
- Dynamic Programming for storing intermediate results.
- Mathematical Formula (Compound Interest) for direct computation in constant time when recursion is unnecessary.

5. Conclusion

The Financial Forecasting tool was successfully implemented using recursion. The recursive approach demonstrates how future financial values can be predicted based on past growth rates. Although recursion is easy to understand and implement, optimization techniques such as memoization or iterative computation can improve performance for larger inputs.